

Manual Statistics Task

Dataset ÷ 78, 85, 92, 88, 95, 73, 81, 90, 87, 93, 76, 89

Mean

$$\text{formula} \rightarrow \bar{X} = \frac{\sum x}{n}$$

$$\sum x = 78 + 85 + 92 + 88 + 95 + 73 + 81 + 90 + 87 + 93 + 76 + 89$$

$$= \underline{\underline{1027}}$$

$$n = 12$$

$$\therefore \text{mean} = \frac{1027}{12} = \underline{\underline{85.58}}$$

Median

arrange data in ascending order

73, 76, 81, 85, 87, 88, 89, 90, 92, 93, 95

no. of observations = 12

median = average of 6th and 7th terms.

$$\text{median} = \frac{87 + 88}{2} = \underline{\underline{87.5}}$$

Variance

$$\text{formula} \rightarrow \sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{mean} = 85.58$$

X	$X - 85.58$	$(X - 85.58)^2$
78	-7.58	57.46
85	-0.58	0.34
92	6.42	41.22
88	2.42	5.86
95	9.42	88.74
73	-12.58	158.26
81	-4.58	20.98
90	4.42	19.54
87	1.42	2.02
93	7.42	55.06
76	-9.58	91.78
89	3.42	11.70

$$\sum (X - \bar{X})^2 = 552.96$$

$$\text{Variance} = \frac{552.96}{12} = \underline{\underline{46.08}}$$

Standard deviation

$$\begin{aligned}\text{formula} \rightarrow \sigma &= \sqrt{\sigma^2} \\ &= \sqrt{46.08} \\ &= 6.78 \\ &= \underline{\underline{6.78}}\end{aligned}$$

Probability of score ~~grd~~ greater than 90

Scores $> 90 \div 92, 95, 93$

no: of favourable outcomes = 3

Total outcomes = 12

~~formula~~

$$\text{formula} \rightarrow P(X > 90) = \frac{\text{favourable outcomes}}{\text{Total outcomes}}$$

$$\text{i.e., } P(X > 90) = \frac{3}{12}$$

$$= 0.25$$

$$\text{i.e., } 25\% //$$

probability of score greater than 90 = 25%